

Day 6 Homework: a chatbot over my own documents

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1. My 20 documents

I used 20 public help pages from the Ollama project (they are free to use under the MIT licence). I put them in **data/my_docs** and built the index. It made **394 chunks** in total, and every file produced at least one chunk.

File	Chunks	File	Chunks
ollama_api.md	135	ollama_import.md	7
ollama_faq.md	45	ollama_docker.md	7
ollama_modelfile.md	33	ollama_cloud.md	6
ollama_gpu.md	30	ollama_cli.md	5
ollama_troubleshooting.md	20	ollama_macos.md	5
ollama_quickstart.md	19	ollama_context-length.md	4
ollama_development.md	15	ollama_index.md	4
ollama_template.md	15	ollama_examples.md	1
ollama_cap_structured-outputs.md	13	ollama_cap_embeddings.md	8
ollama_windows.md	12	ollama_linux.md	10

Total: 394 chunks from 20 documents.

2. My 10 test questions

Seven of the questions can be answered from the documents (four use the document's own words and three use different words). Three cannot (two are not covered at all and one is close but not covered). They are in **data/my_test_questions.csv**.

#	Question	Expected answer	Should it...
1	What port does Ollama listen on by default?	127.0.0.1, port 11434	answer
2	How long does Ollama keep a model in memory by default before unloading it?	5 minutes	answer
3	Which environment variable sets a different directory for storing models?	OLLAMA_MODELS	answer
4	What is the default value of the temperature parameter in a Modelfile?	0.8	answer

5	What command frees a model from memory straight away?	ollama stop <model>	answer
6	How can I make Ollama reachable from other computers on my network?	Set the OLLAMA_HOST variable to change the bind address	answer
7	Where do I find Ollama's error history on Windows?	%LOCALAPPDATA%\Ollama (server.log)	answer
8	Can I run Ollama on a Raspberry Pi?	Not in the documents	decline
9	Does Ollama run on FreeBSD?	Not in the documents	decline
10	How much does an Ollama cloud subscription cost per month?	Not in the documents	decline

3. Results

Run	RAG_MIN_SCORE	Behaviour correct (the script's score)
Before	0.50	7 of 10
After	0.60	7 of 10 (the answers did not change)

I kept the threshold at **0.50**, because raising it did nothing.

Why the threshold did not help

The questions the documents cannot answer still found chunks that scored 0.67 to 0.78. The one answerable question I looked at (question 6) scored 0.77. They score about the same, so no single cut-off can tell them apart.

The script's score is not quite right

On questions 6, 8, 9 and 10 the bot said "I don't know: the documents don't cover that" and then added a sentence of explanation. The script only counts a decline if the reply is exactly that one sentence. So it marked questions 8, 9 and 10 as wrong when the bot had declined correctly, and it marked question 6 as right when the bot was actually wrong. When I read the answers myself, 9 of 10 were right. Only question 6 was wrong.

#	What the script said	What I found when I read the answer
1 to 5, 7	OK	Right answer, with a source
6	OK	Wrong. The bot said the documents don't cover it, but the answer is in ollama_faq.md
8, 9, 10	Wrong	Right. The bot declined, with an extra explanation

4. What I found

Documents

20 Ollama help pages, 394 chunks, no file with 0 chunks.

Results

7 of 10 at 0.50 and 7 of 10 at 0.60. I kept 0.50.

Failures

- **Wrong answer:** none among the six questions it answered.
- **Wrong source:** none that I found. I checked the text of the sources for questions 1, 4 and 7.
- **Answered when it should decline:** none in reality. Questions 8, 9 and 10 were marked wrong only because of the script's exact-match rule.
- **Declined when it should answer:** question 6 ("How can I make Ollama reachable from other computers on my network?"). The FAQ says to change the bind address with the OLLAMA_HOST variable, and the bot still said it did not know.

What I want fixed next (Day 7)

Question 6. I want to look at how the text is cut into chunks, and change the decline check so that a reply that starts with the decline sentence counts as a decline.

Rule added to CLAUDE.md

Chatbot answers must cite their sources. Questions the documents don't cover must be declined. Run the test questions before and after changing the threshold, the chunk size or the number of chunks retrieved, and record both scores.

5. Reflection

Where a chatbot like this would help most

Answering questions that come up again and again and are already written down in a runbook, user guide or policy, such as an IT support or onboarding desk. It worked best here on questions with a simple fact in one place, like the default port, the 5-minute unload time and the OLLAMA_MODELS setting.

The one risk I would check first

That it says "the documents don't cover that" when they do. Question 6 did exactly this. A wrong "I don't know" is easy to miss, because there is no source to check, and the person may stop looking. I would first run questions I know the documents answer and count how many it wrongly declines.

What I would change in my test questions

They were too easy near the edge. The cloud-price question was declined without trouble. The Raspberry Pi question is a better "close but not covered" one, because the Linux page has an ARM64 install but never mentions the Pi.